

Project Design Phase-II
Data Flow Diagram & User Stories

Date	1 March 2026
Team ID	699d5f82fdf409fdbd1bec66
Project Name	Malaria Detection using Deep Learning
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) shows how the data moves through the Malaria Detection System. It illustrates how users upload blood smear images, how the system processes them using a trained deep learning model, and how the prediction results are returned.

The system mainly consists of the following components:

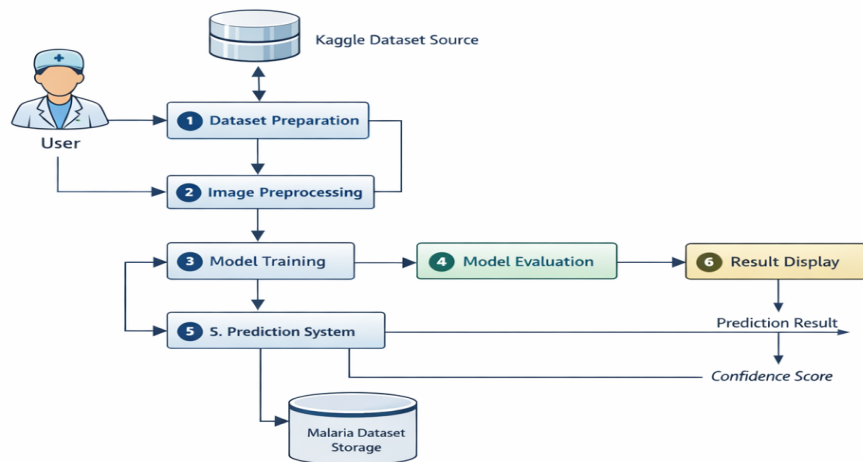
- User Interface (Flask Web Application)
- Image Preprocessing Module
- Deep Learning Prediction Model
- Result Generation Module
- Dataset Storage

The input to the system is a microscopic blood smear image, and the output is the prediction result indicating whether the cell is Parasitized or Uninfected.

Level 0 DFD: Malaria Detection System (Context Diagram)



Level 1 DFD: Malaria Detection System



Level 0 DFD (Context Diagram)

The Level 0 Data Flow Diagram represents the overall interaction between the user and the Malaria Detection System. In this diagram, the system is shown as a single process that interacts with the external user. The user uploads a microscopic blood smear image through the web interface. The system processes the uploaded image using the trained deep learning model and returns the prediction result to the user. The output includes the classification of the blood cell as Parasitized or Uninfected along with the confidence score.

This diagram provides a high-level overview of how data flows between the user and the malaria detection system.

Level 1 DFD

The Level 1 Data Flow Diagram provides a more detailed view of the internal processes of the Malaria Detection System. In this level, the main system is divided into multiple processes such as image upload, image preprocessing, malaria detection, and result display.

First, the user uploads a blood smear image through the web interface. The system then performs preprocessing operations including image resizing, RGB conversion, and normalization. The pre-processed image is passed to the trained deep learning model for analysis. The model predicts whether the blood cell is Parasitized or Uninfected. Finally, the system displays the prediction result along with the confidence score to the user.

The Level 1 DFD illustrates the detailed flow of data within the system components.

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Lab Technician / Healthcare User	Image Upload	USN-1	As a healthcare user, I want to upload a microscopic blood smear image so that the system can detect malaria infection.	I can access my account / dashboard	High	Sprint-1
Lab Technician / Healthcare User	Image Validation	USN-2	As a healthcare user, I want the system to validate the uploaded image before processing.	I can receive confirmation email & click confirm	High	Sprint-1

Lab Technician / Healthcare User	Prediction Processing	USN-3	As a healthcare user, I want the system to analyse the uploaded image using the trained model.	I can register & access the dashboard with Facebook Login	High	Sprint-1
Lab Technician / Healthcare User	Result Display	USN-4	As a healthcare user, I want to see whether the blood cell is infected or not.	System should display prediction result as Parasitized or Uninfected.	High	Sprint-1
Lab Technician / Healthcare User	Confidence Score	USN-5	As a healthcare user, I want to view the prediction confidence so that I can trust the result.	System should display prediction probability or confidence percentage.	Medium	Sprint-2
Lab Technician / Healthcare User	Processing Time	USN-6	As a healthcare user, I want to see how long the prediction took.	System should display processing time along with the prediction result.	Low	Sprint-2
System Developer	Dataset Download	USN-7	As a developer, I want the system to download the malaria dataset automatically from Kaggle.	Dataset should download successfully and be stored locally.	High	Sprint-1
System Developer	Dataset Preparation	USN-8	As a developer, I want the system to split the dataset into training and validation sets.	Dataset should be split with an 80/20 ratio for training and validation.	High	Sprint-1
System Developer	Data Preprocessing	USN-9	As a developer, I want to preprocess images before training the model.	Images should be resized, normalized, and augmented.	High	Sprint-1
System Developer	Model Training	USN-10	As a developer, I want to train deep learning models to classify malaria images.	Models such as Custom CNN, MobileNetV2, and EfficientNetB0 should train successfully.	High	Sprint-2
System Developer	Model Evaluation	USN-11	As a developer, I want to evaluate model performance using metrics.	System should compute accuracy, precision, recall, F1 score, and ROC-AUC.	High	Sprint-2

System Developer	Best Model Selection	USN-12	As a developer, I want the system to select the best performing model automatically.	The model with the highest ROC-AUC and accuracy should be selected.	Medium	Sprint-2
System Developer	Model Export	USN-13	As a developer, I want to export the best trained model for deployment.	Best model should be saved as malaria_detector_final.h5.	Medium	Sprint-2
System Developer	Model Deployment	USN-14	As an administrator, I want to deploy the trained model through a web application.	Flask application should load the trained model successfully.	High	Sprint-3
System Developer	System Health Monitoring	USN-15	As an administrator, I want to check the system health status.	System should provide a health endpoint indicating if the model and server are running.	Low	Sprint-3